Matrix operations: solutions

28-E01 Two triangular substitutions

Derivation

Multiplier $m_{21}=3$ gives $L=[1\ 0;3\ 1]$, $U=[1\ 2;0\ -2]$, $P=I$. Forward $Ly=b$ gives $y_1=5$, $y_2=11-3\cdot 5=-4$. Backward $Ux=y$ gives $x_2=-4/-2=2$, $x_1=5-2\cdot 2=1$. Original rows give $1+4=5$ and $3+8=11$.

Matrix$[0\ 1;1\ 0]$ squares to I, so it is its own inverse. Its first zero pivot invalidates that unpermuted elimination order, not invertibility. Swapping its rows gives $PA=I$ and $L=U=I$.

Conclusion and checks

Solution $x=(1,2)^T$ follows invariants on solved rows. A zero diagonal alone is not a singularity certificate; LUP's test uses an all-zero active pivot column.

28-E02 Schur complement and inverse

Derivation

Here $B=1$, $C=1$, $D=2$, $E=5$, so $S=5-2\cdot 1\cdot 1=3$. Formula entries are $1+1\cdot 1\cdot (1/3)\cdot 2\cdot 1=5/3$, $-1\cdot 1\cdot (1/3)=-1/3$, $-(1/3)\cdot 2\cdot 1=-2/3$, and $1/3$. Thus $A^{-1}=[5/3\ -1/3;-2/3\ 1/3]$.

Product entries are $5/3-2/3=1$, $-1/3+1/3=0$, $10/3-10/3=0$, $-2/3+5/3=1$. Singular B has no B^{-1}, so the formula cannot be applied in hopes of cancellation. For invertible $A=[0\ 1;1\ 0]$, $B=0$ requires another permutation or partition.

Conclusion and checks

$S=3$, $\det A=3$, and the stated inverse checks. Scalar blocks commute here, but general block matrices retain factor order.

28-E03 A line fit with nonzero residual

Derivation

$A=[1\ 0;1\ 1;1\ 2]$, $b=(0,2,1)^T$, $A^TA=[3\ 3;3\ 5]$, $A^Tb=(3,4)^T$. Subtraction gives $2m=1$, then $3c+3/2=3$, so $c=m=1/2$. Predictions $p=(1/2,1,3/2)$, residual $r=(-1/2,1,-1/2)$.

$A^Tr=(-1/2+1-1/2,0\cdot(-1/2)+1\cdot 1+2\cdot(-1/2))=(0,0)$, and squared norm $1/4+1+1/4=3/2$. These are vertical y-errors at fixed t, not shortest scatterpoint-to-line distances.

Matrix operations: solutions (continued)

Changing b to(1,2,3) leaves A unchanged, with independent columns and rank 2. New unique coefficients are c=m=1, residual 0. Data collinearity determines exact fit to this model, not rank of the design formed from t and basis functions.

Conclusion and checks

Minimum error changes from 3/2 to 0 while Gram stays identical with determinant 6>0. Nonzero residual and coefficient uniqueness have different conditions.

28-E04 Later permutations must swap earlier L entries

Derivation

First-column largest magnitude 2 swaps rows 1,3, so p=(3,2,1). Multipliers 1/2 and 0 leave active block[−1/2 −2;2 1]. Second-column largest magnitude 2 swaps current rows 2,3, so p=(3,1,2). Saved first-column multipliers become 0 in row 2 and 1/2 in row 3. New m_{32}=−1/4 and last pivot−7/4.

Final L=[1 0 0;0 1 0;1/2 −1/4 1], U=[2 3 4;0 2 1;0 0 −7/4]. Product rows[2,3,4],[0,2,1], [1,1,0] equal PA in original order 3,1,2.

Pb=(7,0,0), forward y=(7,0,−7/2), backward x=(1,−1,2). Original A gives(0,0,7). Incorrectly leaving L_{21}=1/2 makes product row 2 equal(1/2)(2,3,4)+(0,2,1)=(1,7/2,3), instead of PA row 2=(0,2,1).

Conclusion and checks

Both swaps include prior multipliers and synchronize p, Pb. Product of U diagonals is 2・2・(−7/4)=−7. Two swap signs cancel, giving original det A=−7.

28-E05 Both complexity reductions

Derivation

Multiplying T by the proposed inverse gives identity diagonal blocks. Block(1,2) is−B+B=0,(2,3) is−C+C=0, and(1,3) is BC+B(−C)+0=0. Lower blocks are 0. The upper-right inverse block is BC, without ever commuting to CB.

Construction writes $\Theta(n^2)$ entries, absorbed only with $I(n)=\Omega(n^2)$. The inverse call has order 3n, so I(3n)=O(I(n)) is needed to obtain O(I(n)). Arbitrary irregular cost functions do not permit omitting this assumption.

Matrix operations: solutions (continued)

In reverse, levelℓ has 2^ℓ subproblems, each with M(n/2^ℓ)$\leq$c^ℓM(n). Its work is O((2c)^ℓM(n));2c<1 makes the sum O(M(n)), including leaves. For SPD, leading and Schur blocks permit both inversions. For general real invertible A, Gram G=A^TA is SPD, and $G^{-1}A^TA=I$ proves $A^{-1}=G^{-1}A^T$. Extra products cost O(M(n)).

Conclusion and checks

The reductions specify dimension and exact-arithmetic regularity. Forming Gram squares conditioning, so the theoretical reduction is not a default floating-point solver.

28-E06 SPD and Cholesky

Derivation

Compute $C_{11}=\sqrt{4}=2$, $C_{21}=C_{31}=2/2=1$, $C_{22}=\sqrt{(5-1)}=2$, $C_{32}=(1-1\cdot 1)/2=0$, $C_{33}=\sqrt{(3-1^2-0^2)}=\sqrt{2}$. Thus C=[2 0 0;1 2 0;1 0 $\sqrt{2}$]. Product CC^T has rows[4,2,2],[2,5,1], [2,1,3], equal to A. Since C is invertible, every $z\neq 0$ has $z^TAz=||C^Tz||^2>0$.

After pivot 4, the Schur complement is

$$\begin{bmatrix}5 & 1\\ 1 & 3\end{bmatrix}-\begin{bmatrix}2\\ 2\end{bmatrix}\frac{1}{4}\begin{bmatrix}2 & 2\end{bmatrix}=\begin{bmatrix}4 & 0\\ 0 & 2\end{bmatrix}.$$

Its quadratic form $4u^2+2v^2$ is positive for nonzero(u, v), giving later positive pivots 4,2. By contrast, B=[1 2;2 1] has positive diagonals but z=(1,−1) gives $z^TBz=1-2-2+1=-2<0$. Its second elimination pivot is $1-2\cdot 2=-3$.

Conclusion and checks

SPD quantifies every nonzero vector. Original diagonals alone do not suffice. Here SPD Schur complements make every Cholesky step legal without row pivoting.

28-E07 Rank deficiency and error spaces

Derivation

Every prediction is s(1,1,1), where $s=c_1+c_2$. Squared error is $s^2+(2-s)^2+(1-s)^2=3(s-1)^2+2$, minimized at s=1. All coefficients are(t,1−t), unique prediction(1,1,1), residual(−1,1,0), squared norm 2, and $A^Tr=(0,0)$.

A nullspace coefficient change preserves s and error. Changing the sum to $1+\delta$ gives $2+3\delta^2$, strictly larger for $\delta\neq 0$. Rank is 1, and normal equations give only $c_1+c_2=1$; Gram is not invertible.

Matrix operations: solutions (continued)

Matrix diag(1,10⁻³) has singular values 1,10⁻³, so $\kappa_2=10^3$. Gram diag(1,10⁻⁶) has condition 10⁶. This measures possible directional perturbation amplification, not 10⁶ executed operations. Factorization still has a cubic arithmetic bound in dimension n.

Conclusion and checks

Coefficients may be nonunique while prediction and orthogonal residual are unique. Sensitivity, uniqueness, and operation complexity require separate criteria.

28-E08 One vector in two coordinate systems

Compute $\|p\|^2 = 7/2$, $\|r\|^2 = 3/2$, and $p^T r = -1/4 + 1 - 3/4 = 0$. Thus

$$u = \frac{(1,2,3)^T}{\sqrt{14}}, \qquad v = \frac{(-1,2,-1)^T}{\sqrt{6}}.$$

Both have unit norm and their inner product is zero. Since b=p+r,

$$[b]_{(u,v)} = (u^T b, v^T b)^T = \left(\sqrt{7/2}, \sqrt{3/2}\right)^T.$$

Reconstruction gives $\sqrt{7/2}\,u + \sqrt{3/2}\,v = p + r = (0,2,1)^T$. Two section coordinates and three original coordinates describe the same vector; their component lists are not interchangeable.

Because p=A(1/2,1/2)ᵀ, span{p} lies in E∩Col(A). Conversely, if z=αp+βr belongs to Col(A), then $A^T r=0$ gives $0=r^T z=\beta\|r\|^2$. Hence β=0 and z belongs to span{p}. This proves equality of the subspaces. A has two independent columns, so Col(A) is still two-dimensional. The picture shows its intersection with another two-dimensional plane.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

Matrix operations: solutions (continued)

Verification code appendix: least_squares_check.py

```
"""Check the exact least-squares example and its orthogonality condition."""


def mat_vec(a, v):
    return [sum(row[j] * v[j] for j in range(len(v))) for row in a]


def dot(u, v):
    return sum(x * y for x, y in zip(u, v))


def solve_2x2(a, b):
    det = a[0][0] * a[1][1] - a[0][1] * a[1][0]
    assert det != 0
    return [
        (b[0] * a[1][1] - a[0][1] * b[1]) / det,
        (a[0][0] * b[1] - b[0] * a[1][0]) / det,
    ]


def run():
    a = [[1, 0], [1, 1], [1, 2]]
    b = [0, 2, 1]
    ata = [[dot([row[i] for row in a], [row[j] for row in a]) for j in range(2)] for i in
range(2)]
    atb = [dot([row[i] for row in a], b) for i in range(2)]
    assert ata == [[3, 3], [3, 5]]
    assert atb == [3, 4]
    solution = solve_2x2(ata, atb)
    assert solution == [0.5, 0.5]
    prediction = mat_vec(a, solution)
    residual = [target - fitted for target, fitted in zip(b, prediction)]
    assert prediction == [0.5, 1.0, 1.5]
    assert residual == [-0.5, 1.0, -0.5]
    assert [dot([row[i] for row in a], residual) for i in range(2)] == [0, 0]
    assert round(dot(residual, residual), 12) == 1.5
    print("PASS least-squares intercept=0.5 slope=0.5 residual=(-0.5,1,-0.5) AtR=0")
    return solution, residual


if __name__ == "__main__":
    run()
```

Matrix operations: solutions (continued)

Verification code appendix: matrix_operations_checks.py

```
"""Exact rational matrix identities, independently checked small determinants."""
from fractions import Fraction as F
from itertools import product,permutations
from math import prod
import random

SAMPLE=[[0,2,1],[1,1,0],[2,3,4]]


def transpose(a):return [list(row) for row in zip(*a)]
def identity(n):return [[F(i==j) for j in range(n)] for i in range(n)]
def multiply(a,b):return [[sum((x*y for x,y in zip(row,col)),F(0)) for col in zip(*b)] for
row in a]
def matvec(a,x):return [sum((v*w for v,w in zip(row,x)),F(0)) for row in a]
def subtract(a,b):return [[x-y for x,y in zip(r,s)] for r,s in zip(a,b)]


def determinant(a):
    n=len(a);answer=F(0)
    for p in permutations(range(n)):
        inversions=sum(p[i]>p[j] for i in range(n) for j in range(i+1,n))
        answer+=(-1)**inversions*prod((a[i][p[i]] for i in range(n)),start=F(1))
    return answer


def lup(a):
    n=len(a);work=[[F(x) for x in row] for row in a];order=list(range(n));trace=[];swaps=0
    for k in range(n):
        pivot=max(range(k,n),key=lambda i:abs(work[i][k]))
        if work[pivot][k]==0:raise ValueError('singular matrix')
        if pivot!=k:
            work[k],work[pivot]=work[pivot],work[k]
            order[k],order[pivot]=order[pivot],order[k];swaps+=1
        for i in range(k+1,n):
            work[i][k]/=work[k][k]
            for j in range(k+1,n):work[i][j]-=work[i][k]*work[k][j]
        trace.append(([row[:] for row in work],order[:]))
    lower=[[F(i==j) if i<=j else work[i][j] for j in range(n)] for i in range(n)]
```

Matrix operations: solutions (continued)

```
    upper=[[work[i][j] if i<=j else F(0) for j in range(n)] for i in range(n)]
    return lower,upper,order,trace,swaps


def solve(factor,b):
    lower,upper,order,_,_=factor;n=len(b);y=[F(0)]*n;x=y[:]
    for i in range(n):y[i]=F(b[order[i]])-sum((lower[i][j]*y[j] for j in range(i)),F(0))
    for i in range(n-1,-1,-1):x[i]=(y[i]-sum((upper[i][j]*x[j] for j in
range(i+1,n)),F(0)))/upper[i][i]
    return x,y


def inverse(a):
    n=len(a);factor=lup(a);columns=[solve(factor,[int(i==j) for i in range(n)])[0] for j in
range(n)]
    return transpose(columns)


def ldlt(a):
    n=len(a);lower=identity(n);diagonal=[F(0)]*n
    assert a==transpose(a)
    for j in range(n):
        diagonal[j]=a[j][j]-sum((lower[j][k]**2*diagonal[k] for k in range(j)),F(0))
        if diagonal[j]<=0:raise ValueError('not positive definite')
        for i in range(j+1,n):
            lower[i][j]=(a[i][j]-sum((lower[i][k]*lower[j][k]*diagonal[k] for k in
range(j)),F(0)))/diagonal[j]
    return lower,[[diagonal[i] if i==j else F(0) for j in range(n)] for i in range(n)]


def schur(a,cut):
    b=[row[:cut] for row in a[:cut]];c=[row[cut:] for row in a[:cut]]
    d=[row[:cut] for row in a[cut:]];e=[row[cut:] for row in a[cut:]]
    return subtract(e,multiply(multiply(d,inverse(b)),c))


def least_squares(a,b):
    at=transpose(a);gram=multiply(at,a);rhs=matvec(at,b)
    x,y=solve(lup(gram),rhs);prediction=matvec(a,x);residual=[F(u)-v for u,v in
zip(b,prediction)]
```

Matrix operations: solutions (continued)

```
    return x,prediction,residual,gram,rhs


def main():
    count=0;regular=0;inversions=0
    for n in range(1,4):
        for values in product([-1,0,1],repeat=n*n):
            a=[list(values[i*n:(i+1)*n]) for i in range(n)];det=determinant(a)
            try:factor=lup(a)
            except ValueError:
                assert det==0;count+=1;continue
            assert det!=0;regular+=1
            l,u,p,trace,swaps=factor
            assert multiply(l,u)==[a[i] for i in p]
            assert (-1)**swaps*prod((u[i][i] for i in range(n)),start=F(1))==det
            expected=[F(i+1) for i in range(n)];b=matvec(a,expected)
            assert solve(factor,b)[0]==expected
            if count%17==0 or n<3:
                inv=inverse(a);assert
multiply(a,inv)==identity(n)==multiply(inv,a);inversions+=1
            count+=1
    factor=lup(SAMPLE);l,u,p,trace,swaps=factor
    assert p==[2,0,1] and swaps==2
    assert l==[[1,0,0],[0,1,0],[F(1,2),F(-1,4),1]]
    assert u==[[2,3,4],[0,2,1],[0,0,F(-7,4)]]
    assert solve(factor,[0,0,7])==([1,-1,2],[7,0,F(-7,2)])
    rng=random.Random(28);spd_cases=0;least_cases=0
    for n in range(1,7):
        for _ in range(40):
            c=[[F(rng.randrange(-3,4)) for _ in range(n)] for _ in range(n)]
            a=multiply(transpose(c),c)
            for i in range(n):a[i][i]+=1
            lower,diag=ldlt(a);assert multiply(multiply(lower,diag),transpose(lower))==a
            if n>1:
                for cut in range(1,n):
                    s=schur(a,cut);ls,ds=ldlt(s);assert
multiply(multiply(ls,ds),transpose(ls))==s
            spd_cases+=1
    for m in range(2,8):
```

Matrix operations: solutions (continued)

```
        for n in range(1,min(4,m+1)):
            for _ in range(12):
                a=[[F(rng.randrange(-3,4)) for _ in range(n)] for _ in range(m)]
                b=[F(rng.randrange(-3,4)) for _ in range(m)]
                try:x,prediction,residual,gram,rhs=least_squares(a,b)
                except ValueError:continue
                assert matvec(transpose(a),residual)==[0]*n
                base=sum(v*v for v in residual)
                for direction in product([-1,0,1],repeat=n):
                    changed=[v+w for v,w in zip(x,direction)]
                    updated=[y-z for y,z in zip(b,matvec(a,changed))]
                    assert sum(v*v for v in updated)==base+sum(v*v for v in
matvec(a,direction))
                least_cases+=1
    a=[[1,0],[1,1],[1,2]];b=[0,2,1]
    assert least_squares(a,b)==([F(1,2),F(1,2)],[F(1,2),1,F(3,2)],
                                [F(-1,2),1,F(-1,2)],[[3,3],[3,5]],[3,4])
    plus=multiply(inverse(multiply(transpose(a),a)),transpose(a))
    assert plus==[[F(5,6),F(1,3),F(-1,6)],[F(-1,2),0,F(1,2)]]
    projection=multiply(a,plus)
    assert multiply(plus,a)==identity(2)
    assert multiply(projection,projection)==projection==transpose(projection)
    assert matvec(projection,b)==[F(1,2),1,F(3,2)]
    chol_a=[[4,2,2],[2,5,1],[2,1,3]]
    assert ldlt(chol_a)==([[1,0,0],[F(1,2),1,0],[F(1,2),0,1]],[[4,0,0],[0,4,0],[0,0,2]])
    assert inverse([[1,1],[2,5]])==[[F(5,3),F(-1,3)],[F(-2,3),F(1,3)]]
    # Without swapping the already-built L prefix, the sample factorization fails.
    wrong_l=[row[:] for row in l];wrong_l[1][0],wrong_l[2][0]=wrong_l[2][0],wrong_l[1][0]
    assert multiply(wrong_l,u)!=[SAMPLE[i] for i in p]
    print(f'PASS exact_small_matrices={count} nonsingular={regular}
sampled_two_sided_inverses={inversions} '
          f'SPD_and_all_leading_Schur_cases={spd_cases}
least_squares_orthogonality_and_Pythagoras={least_cases} '
          'L_prefix_swap_counterexample=True sample_x=1,-1,2 residual_squared=3/2')


if __name__=='__main__':main()
```

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study